INTRODUCTION TO INFORMATION GEOMETRY

Frequency-Dependent Fitness and Stability

Week 1 · Lecture 2

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The scalar replicator model

From the first lecture,

$$x' = x(1-x)\Delta(x)$$

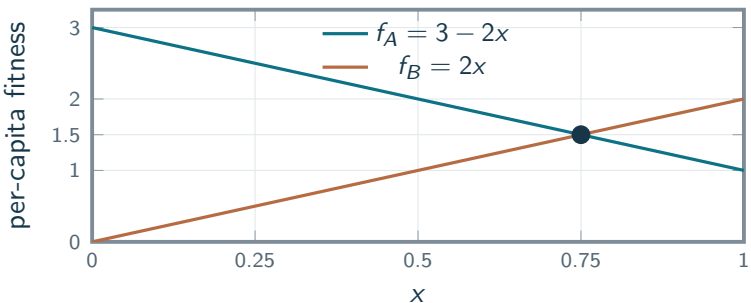
$$\Delta(x) = f_A(x) - f_B(x).$$

The fitness difference may vary with the current proportion. This makes the direction of motion depend on the state.

Fitnesses that depend on frequency

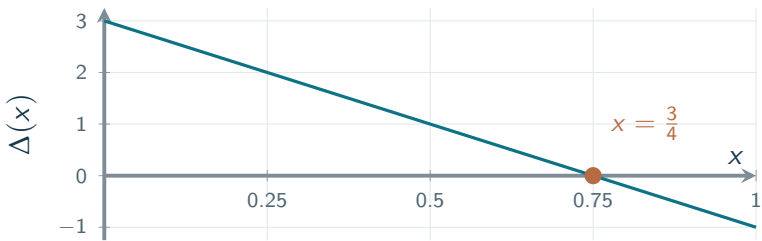
Consider

$$f_A(x) = 3 - 2x, \qquad f_B(x) = 2x.$$



The fitness difference

$$\Delta(x) = (3 - 2x) - 2x = 3 - 4x.$$



Equilibria of the frequency model

The equation becomes

$$x' = F(x) = x(1-x)(3-4x).$$

An equilibrium e satisfies $F(e) = 0$.

$$e = ?$$

Equilibria of the frequency model

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An equilibrium e satisfies $F(e) = 0$.

$$e = 0, \quad \frac{3}{4}, \quad 1.$$

The phase line

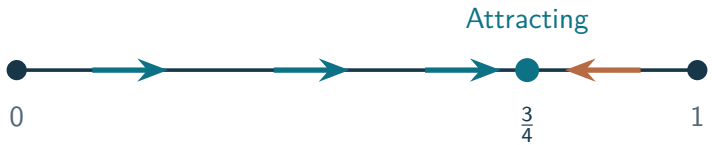
The equilibria are 0, $\frac{3}{4}$, and 1.



Draw the arrows using the sign of $F(x)$. Which equilibrium attracts?

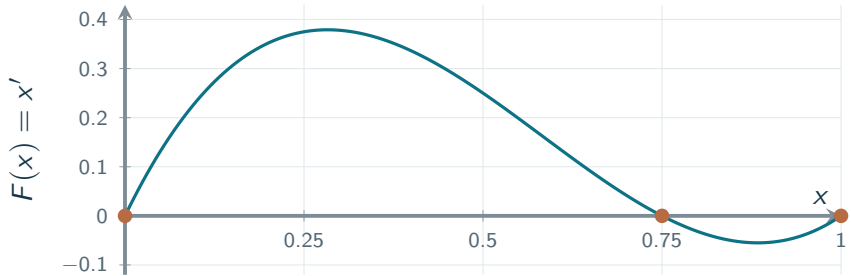
The phase line

The equilibria are 0, $\frac{3}{4}$, and 1.



$F(x) > 0$ for $0 < x < \frac{3}{4}$, $\quad F(x) < 0$ for $\frac{3}{4} < x < 1$.

Velocity as a function of state



The height of the graph gives the slope of a solution at that state.

Reading instantaneous velocities

For

$$F(x) = x(1-x)(3-4x),$$

calculate

$$F(0.4) = ? \qquad \text{and} \qquad F(0.9) = ?.$$

Which solution is increasing? Which is decreasing?

Reading instantaneous velocities

For

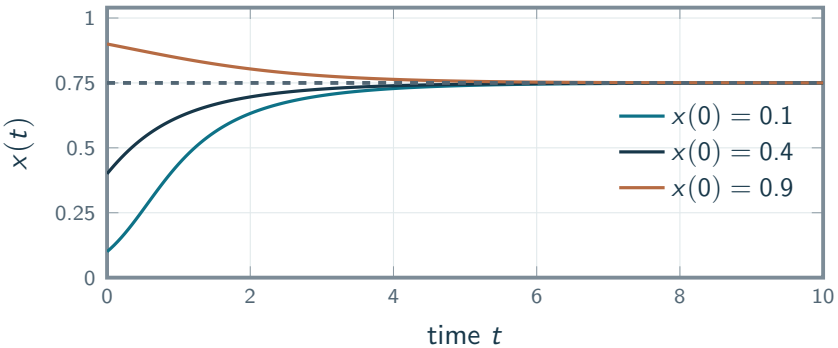
$$F(x) = x(1-x)(3-4x),$$

calculate

$$F(0.4) = 0.336 \qquad \text{and} \qquad F(0.9) = -0.054.$$

The solution at $x = 0.4$ increases; the one at $x = 0.9$ decreases.

Trajectories from three starting states



Every interior trajectory approaches the equilibrium composition $x = \frac{3}{4}$.

Three descriptions of the same motion

1. The phase line records the direction of motion at each state.
2. The graph of $F(x)$ gives the instantaneous velocity at that state.
3. The solution $x(t)$ traces one trajectory through time.

When $x(t) = 0.4$, the time-graph slope must equal

$$x'(t) = F(0.4) = 0.336.$$

Stability on the interval

An equilibrium e is *stable* if solutions starting sufficiently close remain close for all future time.

It is *asymptotically stable* if those nearby solutions also converge to e. For the

coexistence model, the phase-line arrows point toward

$$e = \frac{3}{4}.$$

Linearization near an equilibrium

Let $x = e + u$, where u is a small displacement. Taylor expansion gives

$$F(e+u) = F(e) + F'(e)u + O(u^2).$$

Because $F(e) = 0$, the leading-order equation is

$$u' = F'(e)u.$$

$F'(e) < 0$ predicts decay, while $F'(e) > 0$ predicts growth.

The derivative at coexistence

$$F(x) = x(1-x)(3-4x), \qquad e = \frac{3}{4}.$$

Find

$$F'(e) = ?$$

You may differentiate the expanded polynomial or use the product rule at the equilibrium.

The derivative at coexistence

$$F(x) = x(1-x)(3-4x), \qquad e = \frac{3}{4}.$$

Find

$$F'(e) = \frac{3}{4}\left(1 - \frac{3}{4}\right)(-4) = -\frac{3}{4}.$$

The negative derivative predicts exponential decay of small displacements.

Exponential approach to equilibrium

At $e = \frac{3}{4}$, differentiation gives

$$F'(e) = -\frac{3}{4}.$$

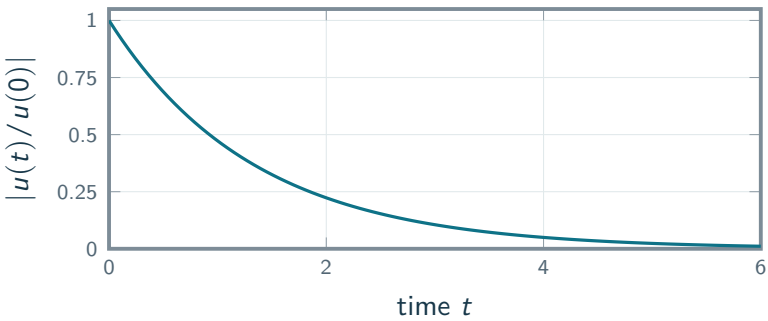
For a small displacement $u = x - \frac{3}{4}$,

$$u(t) \approx u(0)e^{-3t/4}.$$

The displacement decreases by a factor of e in approximately

$$\frac{4}{3} \text{ time units.}$$

The local decay curve



This approximation describes motion close to the equilibrium.

Why solutions remain in $[0, 1]$

$$F(x) = x(1-x)\Delta(x), \quad F(0) = F(1) = 0.$$

Thus $x(t) \equiv 0$ and $x(t) \equiv 1$ are constant solutions. For a smooth fitness difference,

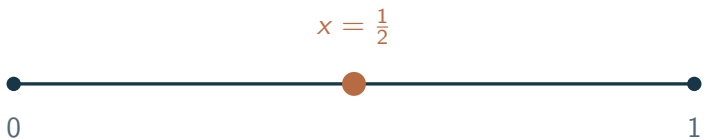
uniqueness of solutions prevents an interior trajectory from crossing either endpoint. The probability interval $[0, 1]$ is forward invariant.

A model with an unstable threshold

Compare the coexistence equation with

$$x' = x(1-x)(4x-2).$$

Its interior equilibrium is $e = \frac{1}{2}$.



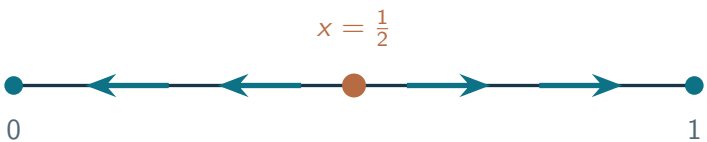
Draw the direction arrows and classify all three equilibria.

A model with an unstable threshold

Compare the coexistence equation with

$$x' = x(1-x)(4x-2).$$

Its interior equilibrium is $e = \frac{1}{2}$.



The endpoints attract; $x = \frac{1}{2}$ is unstable.

A proportion is a probability distribution

Selecting one individual at random gives

$$P = (x, 1 - x)$$



A replicator trajectory is therefore a curve of two-outcome distributions.

Week 1: three mathematical tools

1. A *replicator equation* determines how the proportion changes:

$$x' = x(1-x)(f_A - f_B).$$

2. A *phase line* identifies equilibria and directions of motion.

3. A *linearization* estimates the local rate of approach or departure:

$$u' \approx F'(e)u.$$

Week 2 will obtain the fitness functions from payoff matrices.